

Volume Iii Functions Continuity And Differentiation Reference

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Elementary Functions

0.1 Hyperbolic Functions

Hyperbolic Functions

Definition 0.1.1 (Hyperbolic Functions). For all $x \in \mathbb{R}$:

$$\cosh x := \frac{e^x + e^{-x}}{2}, \quad \sinh x := \frac{e^x - e^{-x}}{2}, \quad \tanh x := \frac{\sinh x}{\cosh x}.$$

Additional: $\operatorname{sech} x := 1/\cosh x$, $\operatorname{csch} x := 1/\sinh x$, $\operatorname{coth} x := \cosh x/\sinh x$.

Proposition 0.1.2 (Hyperbolic Pythagorean Identity). For all $x \in \mathbb{R}$:

$$\cosh^2 x - \sinh^2 x = 1.$$

Go to proof.

Proposition 0.1.3 (Addition Formulas for Hyperbolic Functions). For all $x, y \in \mathbb{R}$:

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y, \quad \cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y.$$

Go to proof.

Proposition 0.1.4 (Limits of $\tanh$).

$$\lim_{x \rightarrow +\infty} \tanh x = 1, \quad \lim_{x \rightarrow -\infty} \tanh x = -1.$$

Go to proof.

Definition 0.1.5 (Inverse Hyperbolic Functions). Since $\sinh$ is strictly increasing and bijective $\mathbb{R} \rightarrow \mathbb{R}$, its inverse $\operatorname{arcsinh}$ is defined on all of $\mathbb{R}$. Since $\cosh$ is strictly increasing on $[0, \infty)$ with range $[1, \infty)$, its inverse $\operatorname{arccosh}$ is defined on $[1, \infty)$. Since $\tanh$ is strictly increasing with range $(-1, 1)$, $\operatorname{arctanh}$ is defined on $(-1, 1)$. Explicit logarithmic formulas:

$$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right), \quad x \in \mathbb{R}.$$

$$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2 - 1}\right), \quad x \geq 1.$$

$$\operatorname{arctanh} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1.$$

0.2 The Logarithm

The Logarithm

Definition 0.2.1 (Natural Logarithm). The **natural logarithm** $\ln : (0, \infty) \rightarrow \mathbb{R}$ is the inverse of the exponential function: for $x > 0$,

$$\ln x := \text{the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently, $\ln x$ is defined by the integral

$$\ln x := \int_1^x \frac{1}{t} dt.$$

Proposition 0.2.2 (Fundamental Logarithmic Limit).

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1.$$

Go to proof.

Proposition 0.2.3 (Every Power Dominates the Logarithm). *For every $r > 0$:*

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x^r} = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} x^r \ln x = 0.$$

Go to proof.

0.3 Power Functions and the Exponential

Power Functions and the Exponential

Definition 0.3.1 (Power Function). Let $n \in \mathbb{N}$. The **integer power function** is $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) := x^n$. For $r \in \mathbb{R}$ and $x > 0$, the **real power** is defined by

$$x^r := e^{r \ln x}.$$

Definition 0.3.2 (The Number e). The **Euler number** is

$$e := \sum_{n=0}^{\infty} \frac{1}{n!} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828 \dots$$

Both expressions define the same real number.

Definition 0.3.3 (Exponential Function). The **exponential function** $\exp : \mathbb{R} \rightarrow (0, \infty)$ is defined by the absolutely convergent power series

$$e^x := \exp(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad x \in \mathbb{R}.$$

Proposition 0.3.4 (Fundamental Exponential Limit).

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1.$$

Go to proof.

Proposition 0.3.5 (Compound Interest Limit).

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

Go to proof.

Proposition 0.3.6 (Exponential Dominates Every Power). *For every $r > 0$:*

$$\lim_{x \rightarrow \infty} \frac{x^r}{e^x} = 0.$$

Go to proof.

0.4 Trigonometric Functions

Trigonometric Functions

Definition 0.4.1 (Sine and Cosine via Power Series). The **sine** and **cosine** functions are defined for all $x \in \mathbb{R}$ by the absolutely convergent power series

$$\begin{aligned}\sin x &:= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \\ \cos x &:= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\end{aligned}$$

Proposition 0.4.2 (Pythagorean Identity). *For all $x \in \mathbb{R}$:*

$$\sin^2 x + \cos^2 x = 1.$$

Go to proof.

Proposition 0.4.3 (Addition Formulas). *For all $x, y \in \mathbb{R}$:*

$$\sin(x+y) = \sin x \cos y + \cos x \sin y, \quad \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Go to proof.

Definition 0.4.4 (Tangent, Cotangent, Secant, Cosecant).

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$ and $\sec x$ are defined where $\cos x \neq 0$, i.e., $x \neq \pi/2 + k\pi$ for $k \in \mathbb{Z}$.

Definition 0.4.5 (Inverse Trigonometric Functions). • $\arcsin : [-1, 1] \rightarrow [-\pi/2, \pi/2]$ is the inverse of $\sin|_{[-\pi/2, \pi/2]}$.

- $\arccos : [-1, 1] \rightarrow [0, \pi]$ is the inverse of $\cos|_{[0, \pi]}$.
- $\arctan : \mathbb{R} \rightarrow (-\pi/2, \pi/2)$ is the inverse of $\tan|_{(-\pi/2, \pi/2)}$.

Theorem 0.4.6 (Fundamental Trigonometric Limit).

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

Go to proof.

Proposition 0.4.7 (Second Fundamental Trigonometric Limit).

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

Go to proof.

Proofs

Continuity

0.5 Limits

Limits of Functions

Definition 0.5.1 (ε -Closeness of a Function to a Value). Let $X \subseteq \mathbb{R}$, let $f : X \rightarrow \mathbb{R}$, let $L \in \mathbb{R}$, and let $\varepsilon > 0$. The function f is **ε -close to L** if $|f(x) - L| \leq \varepsilon$ for all $x \in X$.

Definition 0.5.2 (Local δ -Closeness Near a Point). Let $f : X \rightarrow \mathbb{R}$, let $L \in \mathbb{R}$, let x_0 be adherent to X , and let $\varepsilon, \delta > 0$. The function f is **ε -close to L near x_0 within δ** if

$$\forall x \in X \ (|x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon).$$

Definition 0.5.3 (Limit of a Function). Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . A real number L is the **limit of f at c** , written $\lim_{x \rightarrow c} f(x) = L$, if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \ \forall x \in A \ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Go to uniqueness proof.

Definition 0.5.4 (Sequential Definition of Limit). Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . The limit $\lim_{x \rightarrow c} f(x) = L$ in the **sequential sense** holds if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \rightarrow c$, the sequence $(f(x_n))$ converges to L .

Definition 0.5.5 (Neighbourhood Form of Limit). Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . The limit $\lim_{x \rightarrow c} f(x) = L$ in the **neighbourhood sense** holds if for every ε -neighbourhood $V_\varepsilon(L)$ there exists a deleted δ -neighbourhood $V_\delta^*(c)$ such that

$$x \in V_\delta^*(c) \cap A \implies f(x) \in V_\varepsilon(L).$$

Proposition 0.5.6 (Neighbourhood Form of the Limit). *Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . Then $\lim_{x \rightarrow c} f(x) = L$ if and only if for every ε -neighbourhood $V_\varepsilon(L)$ there exists a δ -neighbourhood $V_\delta(c)$ such that $x \in V_\delta^*(c) \cap A$ implies $f(x) \in V_\varepsilon(L)$.*

Go to proof.

Theorem 0.5.7 (Uniqueness of Limits of Functions). *Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . Then f has at most one limit at c .*

Go to proof.

Proposition 0.5.8 (Sequential Criterion for Function Limits). *Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . Then $\lim_{x \rightarrow c} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \rightarrow c$, one has $f(x_n) \rightarrow L$.*

Go to proof.

Corollary 0.5.9 (Three Equivalent Forms of the Function Limit). *Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . The ε - δ form of $\lim_{x \rightarrow c} f(x) = L$, the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*

Go to proof.

Theorem 0.5.10 (Limit Implies Local Boundedness). *Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . If $\lim_{x \rightarrow c} f(x) = L$ for some $L \in \mathbb{R}$, then there exist $\delta, M > 0$ such that $|f(x)| \leq M$ for all $x \in A$ with $0 < |x - c| < \delta$.*

Go to proof.

Theorem 0.5.11 (Bounded Limits). *Let $f : A \rightarrow \mathbb{R}$, let c be a cluster point of A , and let $a, b \in \mathbb{R}$. If $a \leq f(x) \leq b$ for all $x \in A$ with $x \neq c$, and $\lim_{x \rightarrow c} f(x) = L$, then $a \leq L \leq b$.*

Go to proof.

Theorem 0.5.12 (Squeeze Theorem for Function Limits). *Let $f, g, h : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . Suppose $\lim_{x \rightarrow c} f(x) = L = \lim_{x \rightarrow c} g(x) = L$, and suppose there exists $\delta_0 > 0$ such that*

$$\forall x \in A, 0 < |x - c| < \delta_0 \Rightarrow f(x) \leq h(x) \leq g(x).$$

Then $\lim_{x \rightarrow c} h(x) = L$.

Go to proof.

Theorem 0.5.13 (Limits Are Local). *Let $E \subseteq X \subseteq \mathbb{R}$, let x_0 be a cluster point of E , let $f : X \rightarrow \mathbb{R}$, and let $\delta > 0$. Then*

$$\lim_{x \rightarrow x_0; x \in E} f(x) = L \iff \lim_{x \rightarrow x_0; x \in E \cap (x_0 - \delta, x_0 + \delta)} f(x) = L.$$

Go to proof.

Proposition 0.5.14 (Limit of Absolute Value). *Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . If $\lim_{x \rightarrow c} f(x) = L$, then $\lim_{x \rightarrow c} |f(x)| = |L|$.*

Go to proof.

Theorem 0.5.15 (Composition of Limits). *Let $f : A \rightarrow B$, let $g : B \rightarrow \mathbb{R}$, let c_1 be a cluster point of A , and let $c_2 \in \mathbb{R}$ be a cluster point of B . Suppose $\lim_{x \rightarrow c_1} f(x) = c_2$ and $\lim_{y \rightarrow c_2} g(y) = L$. If $c_2 \in B$, assume also that $g(c_2) = L$. Then $\lim_{x \rightarrow c_1} g(f(x)) = L$.*

Go to proof.

Theorem 0.5.16 (Monotone Limit Theorem). *Let $a < b$ and let $f : (a, b) \rightarrow \mathbb{R}$ be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing f and every $x_0 \in (a, b)$:*

$$f(x_0^-) = \sup\{f(x) : a < x < x_0\}, \quad f(x_0^+) = \inf\{f(x) : x_0 < x < b\}.$$

Go to proof.

Proposition 0.5.17 (Cauchy Criterion for Function Limits). *Let $f : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . Then $\lim_{x \rightarrow c} f(x) = L$ exists for some $L \in \mathbb{R}$ if and only if*

$$\forall \text{OutputTolerance}(\varepsilon), \exists \text{InputTolerance}(\delta), \forall x, y \in A : 0 < |x - c| < \delta \wedge 0 < |y - c| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon. \blacksquare$$

Go to proof.

Algebra of Limits

Theorem 0.5.18 (Sum Rule for Function Limits). *Let $f, g : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . If $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, then*

$$\lim_{x \rightarrow c} (f + g)(x) = L + M.$$

Go to proof.

Theorem 0.5.19 (Difference Rule for Function Limits). *Let $f, g : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . If $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, then*

$$\lim_{x \rightarrow c} (f - g)(x) = L - M.$$

Go to proof.

Theorem 0.5.20 (Scalar Multiple Rule for Function Limits). *Let $f : A \rightarrow \mathbb{R}$, let c be a cluster point of A , and let $\alpha \in \mathbb{R}$. If $\lim_{x \rightarrow c} f(x) = L$, then*

$$\lim_{x \rightarrow c} (\alpha f)(x) = \alpha L.$$

Go to proof.

Theorem 0.5.21 (Product Rule for Function Limits). *Let $f, g : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . If $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = M$, then*

$$\lim_{x \rightarrow c} (fg)(x) = LM.$$

Go to proof.

Theorem 0.5.22 (Quotient Rule for Function Limits). *Let $f, g : A \rightarrow \mathbb{R}$ and let c be a cluster point of A . If $\lim_{x \rightarrow c} f(x) = L$, $\lim_{x \rightarrow c} g(x) = M$, and $M \neq 0$, then there exists $\delta_0 > 0$ such that $g(x) \neq 0$ whenever $x \in A$ and $0 < |x - c| < \delta_0$, and*

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Go to proof.

One-Sided Limits

Definition 0.5.23 (Right-Hand Limit). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. A real number L is the **right-hand limit** of f at c if*

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Definition 0.5.24 (Left-Hand Limit). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. A real number L is the **left-hand limit** of f at c if*

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A \ 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Theorem 0.5.25 (Sequential Criterion for Right-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. Then $\lim_{x \rightarrow c^+} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (c, \infty)$ with $x_n \rightarrow c$, one has $f(x_n) \rightarrow L$.*

Go to proof.

Corollary 0.5.26 (Sequential Criterion for Left-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. Then $\lim_{x \rightarrow c^-} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (-\infty, c)$ with $x_n \rightarrow c$, one has $f(x_n) \rightarrow L$.*

Go to proof.

Theorem 0.5.27 (Two-Sided Limit via One-Sided Limits). *Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. If c is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x \rightarrow c} f(x) = L$ if and only if both $\lim_{x \rightarrow c^+} f(x) = L$ and $\lim_{x \rightarrow c^-} f(x) = L$.*

Go to proof.

0.6 Point Continuity

Continuity

Definition 0.6.1 (Relative Neighborhood). Let $E \subseteq \mathbb{R}$ and $a \in E$. A set $U \subseteq \mathbb{R}$ is a neighborhood of a in E if and only if there exists $\delta > 0$ such that $U = E \cap (a - \delta, a + \delta)$. Equivalently, every neighborhood of a in E is a set of the form $U_E(a) := E \cap (a - \delta, a + \delta)$ for some $\delta > 0$.

Definition 0.6.2 (Continuity at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. The function f is **continuous at c** if and only if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A : (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon).$$

Definition 0.6.3 (Continuity at a Point — Neighbourhood Form). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. The function f is continuous at c if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f(A \cap (c - \delta, c + \delta)) \subseteq (f(c) - \varepsilon, f(c) + \varepsilon).$$

Definition 0.6.4 (Continuity at a Point — Sequential Form). Let $A \subseteq \mathbb{R}$, let $c \in A$, and let $f : A \rightarrow \mathbb{R}$. The function f is continuous at c if and only if for every sequence $(x_n)_{n \in \mathbb{N}}$ in A , the convergence $\lim_{n \rightarrow \infty} x_n = c$ implies $\lim_{n \rightarrow \infty} f(x_n) = f(c)$.

Characterization of Discontinuity

Definition 0.6.5 (Point of Discontinuity). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $a \in A$. The function f is discontinuous at a if and only if f is not continuous at a .

Definition 0.6.6 (Sequential Characterization of Discontinuity). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a sequence $(x_n)_{n \in \mathbb{N}} \subseteq A$ such that $x_n \rightarrow c$ while $f(x_n) \not\rightarrow f(c)$.

Definition 0.6.7 (Neighbourhood Characterization of Discontinuity). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a neighborhood V of $f(c)$ such that for every neighborhood U of c relative to A there exists $x \in U \cap A$ with $f(x) \notin V$.

Lemma 0.6.8 (Equivalent Formulations of Discontinuity at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. The following are equivalent:

1. c is a point of discontinuity in the ε - δ sense.
2. c is a sequential point of discontinuity.
3. c is a neighbourhood point of discontinuity.

Go to proof.

Definition 0.6.9 (Types of Discontinuity). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $x_0 \in A$ with f discontinuous at x_0 .

- (i) *Removable* iff $\exists L \in \mathbb{R} : \lim_{x \rightarrow x_0} f(x) = L$ and $L \neq f(x_0)$.
- (ii) *Jump* iff $\exists L_-, L_+ \in \mathbb{R} : \lim_{x \rightarrow x_0^-} f(x) = L_-, \lim_{x \rightarrow x_0^+} f(x) = L_+,$ and $L_- \neq L_+.$
- (iii) *Essential* iff it is not removable.

Oscillation

Definition 0.6.10 (Oscillation on a Set). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $E \subseteq A$. The oscillation of f on E is the extended real number

$$\omega(f; E) := \sup\{|f(x_1) - f(x_2)| : x_1, x_2 \in E\} = \text{diam}(f(E)).$$

Definition 0.6.11 (Oscillation at a Point). Let $E \subseteq \mathbb{R}$, let $f : E \rightarrow \mathbb{R}$, and let $x_0 \in \mathbb{R}$ satisfy $U_\delta(x_0) \cap E \neq \emptyset$ for every $\delta > 0$, where $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$. The oscillation of f at x_0 is the extended real number

$$\omega(f; x_0) = \lim_{\delta \rightarrow 0^+} \omega(f; U_\delta(x_0) \cap E),$$

with $\omega(f; A)$ denoting the oscillation of f on the set A . Because the function $\delta \mapsto \omega(f; U_\delta(x_0) \cap E)$ is nonincreasing, this right-hand limit exists (possibly $+\infty$).

Theorem 0.6.12 (Continuity Characterised by Oscillation). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. Then f is continuous at c if and only if $\omega(f; c) = 0$. Go to proof.

Proposition 0.6.13 (Discontinuity Set via Oscillation). Let $f : E \rightarrow \mathbb{R}$. Then

$$D(f) := \{x \in E : f \text{ is discontinuous at } x\} = \{x \in E : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f; x) \geq 1/n\}.$$

Go to proof.

0.7 Global Theorems

Intermediate Value Theorem

Definition 0.7.1 (Bounded on a Set). Let $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$. The function f is bounded on A if and only if there exists a constant $M > 0$ such that $|f(x)| \leq M$ for every $x \in A$.

Theorem 0.7.2 (Boundedness Theorem). Let $a < b$ and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$. Then there exists $M > 0$ such that $|f(x)| \leq M$ for every $x \in [a, b]$. Go to proof.

Definition 0.7.3 (Absolute Maximum and Absolute Minimum). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. A point $x^* \in A$ is an absolute maximum of f on A if and only if $f(x^*) \geq f(x)$ for every $x \in A$. A point $x_* \in A$ is an absolute minimum of f on A if and only if $f(x_*) \leq f(x)$ for every $x \in A$.

Theorem 0.7.4 (Extreme Value Theorem). Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. There exist points $x_m, x_M \in [a, b]$ such that $f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a, b]$. Equivalently, f attains both an absolute maximum and an absolute minimum on $[a, b]$. Go to proof.

Theorem 0.7.5 (Location of Roots). Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. If either $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in (a, b)$ such that $f(c) = 0$. Go to proof.

Theorem 0.7.6 (Bolzano's Intermediate Value Theorem). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$ be continuous on I , and choose $a, b \in I$ with $f(a) < k < f(b)$. Then there exists $c \in I$ such that $\min\{a, b\} < c < \max\{a, b\}$ and $f(c) = k$. Go to proof.

Theorem 0.7.7 (Image of Closed Bounded Interval Is Closed Bounded Interval). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then $f([a, b]) = [\min_{[a, b]} f, \max_{[a, b]} f]$. Go to proof.

Theorem 0.7.8 (Preservation of Intervals). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be continuous. Then $f(I)$ is an interval. Go to proof.

Theorem 0.7.9 (Darboux Property). Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, and fix $c \in \mathbb{R}$. If $f(x) \neq c$ for every $x \in [a, b]$, then either $f(x) > c$ for all $x \in [a, b]$ or $f(x) < c$ for all $x \in [a, b]$. Go to proof.

0.8 Uniform Continuity

Uniform Continuity

Definition 0.8.1 (Uniform Continuity). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The function f is uniformly continuous on A if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, u \in A$ the implication $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$ holds.

Theorem 0.8.2 (Algebra of Uniform Continuity on a General Domain). (a) For every set $A \subseteq \mathbb{R}$, every pair of functions $f, g : A \rightarrow \mathbb{R}$ that are uniformly continuous on A , and every scalar $c \in \mathbb{R}$, the functions $f + g$, $f - g$, and cf are uniformly continuous on A .

(b) There exist a set $A \subseteq \mathbb{R}$ and uniformly continuous functions $f, g : A \rightarrow \mathbb{R}$ such that the product fg is not uniformly continuous on A .

Go to proof.

Theorem 0.8.3 (Algebra of Uniform Continuity on a Bounded Domain). Let $A \subseteq \mathbb{R}$ be bounded, and let $f, g : A \rightarrow \mathbb{R}$ be uniformly continuous on A .

(a) The product $x \mapsto f(x)g(x)$ is uniformly continuous on A .

(b) If there exists $k > 0$ such that $|g(x)| \geq k$ for all $x \in A$, then the quotient $x \mapsto f(x)/g(x)$ is uniformly continuous on A .

Go to proof.

Proposition 0.8.4 (Sequential Characterization of Uniform Continuity). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. Then f is uniformly continuous on A if and only if for every pair of sequences (x_n) and (y_n) in A with $\lim_{n \rightarrow \infty} (x_n - y_n) = 0$ we have $\lim_{n \rightarrow \infty} (f(x_n) - f(y_n)) = 0$. Go to proof.

Theorem 0.8.5 (Heine–Cantor Theorem). Let $a, b \in \mathbb{R}$ with $a < b$ and let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$. Then f is uniformly continuous on $[a, b]$. Go to proof.

Proposition 0.8.6 (Uniform Continuity Preserves Cauchy Sequences). Let $S \subseteq \mathbb{R}$, let $f : S \rightarrow \mathbb{R}$, and let (x_n) be a sequence in S . If f is uniformly continuous on S and (x_n) is a Cauchy sequence in S , then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$. Go to proof.

Definition 0.8.7 (Lipschitz Condition). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $K > 0$. The function f satisfies the Lipschitz condition on A with constant K if and only if $|f(x) - f(u)| \leq K|x - u|$ for every $x, u \in A$.

Proposition 0.8.8 (Lipschitz Implies Uniform Continuity). Let $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$. If there exists $K \geq 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$, then f is uniformly continuous on A . Go to proof.

Theorem 0.8.9 (Algebra of Lipschitz Functions). Let $A \subseteq \mathbb{R}$ and let $f, g : A \rightarrow \mathbb{R}$ be Lipschitz with constants $K_f, K_g \geq 0$ respectively.

(a) For every $c \in \mathbb{R}$, the function cf is Lipschitz on A with constant $|c|K_f$.

(b) The functions $f + g$ and $f - g$ are Lipschitz on A with constant $K_f + K_g$.

(c) If $B \subseteq \mathbb{R}$, $h : B \rightarrow \mathbb{R}$ is Lipschitz with constant $K_h \geq 0$, and $f(A) \subseteq B$, then $h \circ f$ is Lipschitz on A with constant $K_h K_f$.

(d) If $|f(x)| \leq M_f$ and $|g(x)| \leq M_g$ for all $x \in A$, then fg is Lipschitz on A with constant $M_g K_f + M_f K_g$.

(e) If $|f(x)| \leq M_f$ for all $x \in A$, if $|g(x)| \geq k > 0$ for all $x \in A$, and if g is Lipschitz on A with constant K_g , then f/g is Lipschitz on A with constant $K_f/k + M_f K_g/k^2$.

Go to proof.

Definition 0.8.10 (Bi-Lipschitz Map). Let $A \subseteq \mathbb{R}$ and let $f : A \rightarrow \mathbb{R}$. The map f is bi-Lipschitz on A if and only if there exist constants $0 < \alpha \leq K$ such that for every $x, u \in A$,

$$\alpha|x - u| \leq |f(x) - f(u)| \leq K|x - u|.$$

Any such pair (α, K) is called a bi-Lipschitz constant pair for f on A .

Theorem 0.8.11 (Inverse of a Bi-Lipschitz Map Is Lipschitz). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and suppose there exist constants $\alpha, K > 0$ such that $\alpha|x - y| \leq |f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$. Then f is injective, the inverse mapping $f^{-1} : f(A) \rightarrow A$ is well defined, and

$$\frac{1}{K}|u - v| \leq |f^{-1}(u) - f^{-1}(v)| \leq \frac{1}{\alpha}|u - v| \quad \text{for all } u, v \in f(A).$$

Go to proof.

0.9 Monotone Functions

Monotone Functions and Continuity

Theorem 0.9.1 (One-Sided Limits of Monotone Functions). Let $I \subseteq \mathbb{R}$ be an interval, let c be interior to I , and let $f : I \rightarrow \mathbb{R}$ be increasing. Define

$$L_- := \sup\{f(x) : x \in I, x < c\}, \quad L_+ := \inf\{f(x) : x \in I, c < x\}.$$

Then both one-sided limits exist and satisfy

$$\lim_{x \rightarrow c^-} f(x) = L_-, \quad \lim_{x \rightarrow c^+} f(x) = L_+.$$

Go to proof.

Corollary 0.9.2 (Continuity Criterion for Monotone Functions). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$ be monotone increasing, and let c be an interior point of I . Then f is continuous at c if and only if its one-sided limits satisfy $f(c^-) = f(c) = f(c^+)$, where $f(c^-) = \lim_{x \rightarrow c^-} f(x)$ and $f(c^+) = \lim_{x \rightarrow c^+} f(x)$. Go to proof.

Definition 0.9.3 (Jump of a Function). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$ be increasing, and let c be an interior point of I . If the one-sided limits $f(c^-)$ and $f(c^+)$ exist, the jump of f at c is

$$j_f(c) := f(c^+) - f(c^-).$$

Proposition 0.9.4 (Discontinuities of a Monotone Function Are of First Kind). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$ be monotone, and let $a \in I$ be such that there exist points of I strictly less and strictly greater than a . If f is discontinuous at a , then both one-sided limits $\lim_{x \uparrow a} f(x)$ and $\lim_{x \downarrow a} f(x)$ exist in $\mathbb{R}$. Go to proof.

Corollary 0.9.5 (Jump Intervals Are Disjoint). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be nondecreasing. If $a, b \in I$ are distinct points of discontinuity of f , then the open intervals $(f(a^-), f(a^+))$ and $(f(b^-), f(b^+))$ are disjoint. Go to proof.

Theorem 0.9.6 (Discontinuities of a Monotone Function Are Countable). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be monotone. Then the set $D_f := \{c \in I : f \text{ is discontinuous at } c\}$ is at most countable. Go to proof.

Corollary 0.9.7 (Monotone Function Is Continuous on a Dense Set). Let $f : [a, b] \rightarrow \mathbb{R}$ be monotone. Then the set $\{x \in [a, b] : f \text{ is continuous at } x\}$ is dense in $[a, b]$. Go to proof.

Proposition 0.9.8 (Continuous Injective Is Strictly Monotone). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$. Then f is injective if and only if f is strictly monotone on $[a, b]$. Go to proof.

Theorem 0.9.9 (Continuous Inverse Theorem). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be strictly monotone and continuous on I . Then the inverse function $f^{-1} : f(I) \rightarrow I$ exists, is strictly monotone, and is continuous on $f(I)$. Go to proof.

Limits Superior and Inferior of Functions

Definition 0.9.10 (Limits Superior and Inferior of a Function). Let $f : E \rightarrow \mathbb{R}$ and let x_0 be an accumulation point of E . For each $\delta > 0$ set

$$U(\delta) = \{ f(x) : x \in E, 0 < |x - x_0| < \delta \}.$$

The extended real numbers $\limsup_{x \rightarrow x_0} f(x)$ and $\liminf_{x \rightarrow x_0} f(x)$ are defined by

$$\limsup_{x \rightarrow x_0} f(x) = \inf_{\delta > 0} \sup U(\delta), \quad \liminf_{x \rightarrow x_0} f(x) = \sup_{\delta > 0} \inf U(\delta),$$

where each inner supremum or infimum is taken in $\overline{\mathbb{R}}$ and the outer infimum or supremum ranges over all $\delta > 0$.

Proposition 0.9.11 ($\limsup \geq \liminf$). Let $A \subseteq \mathbb{R}$, let x_0 be a limit point of A , and let $f : A \rightarrow \mathbb{R}$. Then

$$\limsup_{x \rightarrow x_0} f(x) \geq \liminf_{x \rightarrow x_0} f(x).$$

Go to proof.

Proposition 0.9.12 (Limit Exists iff $\limsup = \liminf$). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, let x_0 be a limit point of A with $x_0 \in \mathbb{R}$, and let $L \in \mathbb{R}$. Then $\lim_{x \rightarrow x_0} f(x)$ exists and equals L if and only if $\limsup_{x \rightarrow x_0} f(x) = \liminf_{x \rightarrow x_0} f(x) = L$. *Go to proof.*

0.10 Limits at Infinity

Limits at Infinity

Definition 0.10.1 (Infinite Adherent Points). Let $X \subseteq \mathbb{R}$. The point $+\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x > M$. The point $-\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x < M$.

Definition 0.10.2 (Limit at Infinity). Let $X \subseteq \mathbb{R}$ be such that for every $M \in \mathbb{R}$ there exists $x \in X$ with $x > M$, let $f : X \rightarrow \mathbb{R}$, and let $L \in \mathbb{R}$. The limit of f at $+\infty$ equals L if and only if for every $\varepsilon > 0$ there exists $M \in \mathbb{R}$ such that for every $x \in X$, $x > M$ implies $|f(x) - L| < \varepsilon$.

0.11 Approximation

Approximation of Continuous Functions

Theorem 0.11.1 (Step Function Approximation). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a step function $s_\varepsilon : [a, b] \rightarrow \mathbb{R}$ such that

$$\sup_{x \in [a, b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

Go to proof.

Definition 0.11.2 (Piecewise Linear Function). Let $a, b \in \mathbb{R}$ with $a < b$ and let $g : [a, b] \rightarrow \mathbb{R}$. The function g is piecewise linear if and only if there exist $m \in \mathbb{N}$ with $m \geq 1$ and points $a = x_0 < x_1 < \dots < x_m = b$ such that $g|_{[x_{k-1}, x_k]}$ is affine for every $k \in \{1, \dots, m\}$.

Theorem 0.11.3 (Piecewise Linear Approximation). Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a continuous piecewise linear function $\ell_\varepsilon : [a, b] \rightarrow \mathbb{R}$ such that $\sup_{x \in [a, b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$. *Go to proof.*

Theorem 0.11.4 (Weierstrass Approximation Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. For every $\varepsilon > 0$ there exists a polynomial $p_\varepsilon \in \mathbb{R}[x]$ such that $\|f - p_\varepsilon\|_{\infty, [a, b]} < \varepsilon$. Go to proof.*

Definition 0.11.5 (Bernstein Polynomial). Let $n \in \mathbb{N}$ and let $f : [0, 1] \rightarrow \mathbb{R}$. A function $B_n : [0, 1] \rightarrow \mathbb{R}$ is called the n th Bernstein polynomial of f if and only if, for every $x \in [0, 1]$,

$$B_n(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

Theorem 0.11.6 (Bernstein Approximation Theorem). *Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n denote the n th Bernstein polynomial of f . Then for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that $\|f - B_n\|_\infty < \varepsilon$ for all $n \geq n_\varepsilon$. Go to proof.*

0.12 Gauge

Gauges and Tagged Partitions

Definition 0.12.1 (Partition). Let $a < b$ and set $I = [a, b]$. A finite ordered set $P = \{x_0, \dots, x_n\}$ with $n \in \mathbb{N}$ and $n \geq 1$ is a partition of I if and only if $x_0 = a$, $x_n = b$, and $x_0 < x_1 < \dots < x_n$. In this case the closed subintervals $[x_{i-1}, x_i]$ for $i = 1, \dots, n$ subdivide I .

Definition 0.12.2 (Tagged Partition). Let $a, b \in \mathbb{R}$ with $a < b$. A list $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is a tagged partition of $[a, b]$ if and only if the subintervals determine a partition $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ of $[a, b]$ with $a = x_0 < x_1 < \dots < x_n = b$, and each tag satisfies $t_i \in [x_{i-1}, x_i]$ for $i = 1, \dots, n$. Equivalently, a tagged partition is exactly a partition P of $[a, b]$ together with one tag chosen inside every subinterval of P .

Definition 0.12.3 (Gauge). Let $a, b \in \mathbb{R}$ with $a < b$ and set $I = [a, b]$. A gauge on I is a function $\delta : I \rightarrow (0, \infty)$.

Definition 0.12.4 (δ -Fine Partition). Let $[a, b]$ be a closed interval, let $\delta : [a, b] \rightarrow (0, \infty)$ be a gauge, and let $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$. The tagged partition $\dot{P}$ is δ -fine if and only if for every $i \in \{1, \dots, n\}$ and every $x \in [x_{i-1}, x_i]$ the inequality $|x - t_i| < \delta(t_i)$ holds.

Lemma 0.12.5 (Every Point Is Covered by a Tag). *Let $\dot{P} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ that is δ -fine on $[a, b]$, and let $x \in [a, b]$. Then there exists $i \in \{1, \dots, n\}$ such that $|x - t_i| < \delta(t_i)$. Go to proof.*

Theorem 0.12.6 (Cousin's Theorem). *Every gauge $\delta : [a, b] \rightarrow (0, \infty)$ on a closed interval $[a, b]$ admits a δ -fine tagged partition of $[a, b]$. Go to proof.*

Tagged Partitions and Mesh

Definition 0.12.7 (Mesh of a Partition). Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. A real number m is the *mesh* (or *norm*) of P if and only if

$$m = \max_{1 \leq i \leq n} (x_i - x_{i-1}).$$

The mesh of P is denoted by $\|P\|$ and is the unique m satisfying this equivalence.

Definition 0.12.8 (Refinement). Let $[a, b] \subseteq \mathbb{R}$ and let P and Q be partitions of $[a, b]$. Then Q is a refinement of P if and only if every point of P is also a point of Q ; equivalently, $P \subseteq Q$.

Definition 0.12.9 (Common Refinement). Let $a < b$ and let P and Q be partitions of $[a, b]$. Their common refinement is the partition R obtained by taking the union of their node sets, so $R = P \cup Q$ as sets of points in $[a, b]$.

Proofs

Differentiation

0.13 The Tangent Problem

Definition 0.13.1 (Secant Line). Let $f : A \rightarrow \mathbb{R}$, and let $x_1, x_2 \in A$ with $x_1 \neq x_2$. The *secant line* determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Definition 0.13.2 (Difference Quotient). Let $f : A \rightarrow \mathbb{R}$, let $c \in A$, and let $x \in A$ with $x \neq c$. The *difference quotient* of f from c to x is

$$m_{\text{sec}}(c, x) := \frac{f(x) - f(c)}{x - c}.$$

Definition 0.13.3 (Tangent Line). Let $f : A \rightarrow \mathbb{R}$, let $c \in A$, and suppose that f is differentiable at c . The *tangent line* to the graph of f at the point $(c, f(c))$ is the line through $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

0.14 The Derivative

Definition 0.14.1 (Derivative at a Point). Let $f : D \rightarrow \mathbb{R}$ be a real-valued function and $c \in \text{int}(D)$. We say f is *differentiable at c* with *derivative $L = f'(c)$* if and only if:

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in D \left(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \right).$$

Definition 0.14.2 (Topological Definition of Derivative at a Point). Let $f : D \rightarrow \mathbb{R}$ and $c \in \text{int}(D)$. f is differentiable at c with derivative L if for every neighbourhood V of L , there exists a neighbourhood U of c such that for all $x \in (U \setminus \{c\}) \cap D$:

$$\frac{f(x) - f(c)}{x - c} \in V.$$

Definition 0.14.3 (Sequential Definition of Derivative at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . We say that f is differentiable at c with derivative L if

$$\forall (x_n) \subseteq A \setminus \{c\} \left(x_n \rightarrow c \implies \frac{f(x_n) - f(c)}{x_n - c} \rightarrow L \right).$$

Theorem 0.14.4 (Equivalence of Definitions of the Derivative at a Point). Let $f : A \rightarrow \mathbb{R}$ and $c \in A$ be a limit point of A . The following are equivalent:

1. f is differentiable at c in the ε - δ sense with derivative L .
2. f is differentiable at c in the topological sense with derivative L .
3. f is differentiable at c in the sequential sense with derivative L .

Go to proof.

Proposition 0.14.5 (Derivative: h -Form Equivalence). *Let $f : A \rightarrow \mathbb{R}$ and $c \in \text{int}(A)$. Then f is differentiable at c with derivative L if and only if*

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = L.$$

Go to proof.

0.14.1 First Consequences

Theorem 0.14.6 (Differentiable Implies Continuous). *Let $f : A \rightarrow \mathbb{R}$ and $c \in A$. If f is differentiable at c , then f is continuous at c .*

Go to proof.

Theorem 0.14.7 (Uniqueness of the Derivative). *Let $f : A \rightarrow \mathbb{R}$ and $c \in A$. If f is differentiable at c , its derivative is unique.*

Go to proof. Go to proof.

Proposition 0.14.8 (Derivative of the Identity). *Let $f(x) = x$ for all $x \in \mathbb{R}$. Then $f'(c) = 1$ for every $c \in \mathbb{R}$.*

Go to proof.

Proposition 0.14.9 (Derivative of a Constant). *Let $k \in \mathbb{R}$ and let $f(x) = k$ for all $x \in \mathbb{R}$. Then $f'(c) = 0$ for every $c \in \mathbb{R}$.*

Go to proof.

0.15 One-Sided Derivatives

Definition 0.15.1 (Left-Hand Derivative). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$ be such that c is a left-hand limit point of I . We say that f has a *left-hand derivative* at c if the limit

$$\lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_-(c)$:

$$f'_-(c) := \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c}.$$

Definition 0.15.2 (Right-Hand Derivative). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$ be such that c is a right-hand limit point of I . We say that f has a *right-hand derivative* at c if the limit

$$\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_+(c)$:

$$f'_+(c) := \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}.$$

Theorem 0.15.3 (Differentiability and One-Sided Derivatives). *Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \rightarrow \mathbb{R}$. Then f is differentiable at $c \in I$ if and only if both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

Go to proof.

0.16 Carathéodory and the Chain Rule

0.16.1 Carathéodory Characterization

Definition 0.16.1 (Carathéodory Factorization). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. We say that f admits a *Carathéodory factorization* at c if there exists a function $\phi : I \rightarrow \mathbb{R}$, continuous at c , such that

$$f(x) = f(c) + (x - c)\phi(x) \quad \text{for all } x \in I.$$

In this case, f is differentiable at c and $f'(c) = \phi(c)$.

Theorem 0.16.2 (Carathéodory Characterization of Differentiability). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. The following are equivalent:

- (i) f is differentiable at c .
- (ii) There exists a function $\phi : I \rightarrow \mathbb{R}$, continuous at c , such that

$$f(x) = f(c) + (x - c)\phi(x) \quad \text{for all } x \in I.$$

In this case, $\phi(c) = f'(c)$. Go to proof. Go to proof.

Theorem 0.16.3 (Chain Rule). Let $I, J \subseteq \mathbb{R}$ be intervals, let $f : I \rightarrow \mathbb{R}$ and $g : J \rightarrow \mathbb{R}$, and suppose that $f(I) \subseteq J$. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then the composition $g \circ f$ is differentiable at c , and

$$(g \circ f)'(c) = g'(f(c)) f'(c).$$

Go to proof.

Theorem 0.16.4 (Leibniz Rule). Let $I \subseteq \mathbb{R}$ be an interval, let $x \in I$, and suppose that $f, g : I \rightarrow \mathbb{R}$ are n -times differentiable on a neighbourhood of x . Then fg is n -times differentiable at x , and

$$(fg)^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Go to proof.

Theorem 0.16.5 (Faa di Bruno's Formula). Let $I, J \subseteq \mathbb{R}$ be intervals, let $x \in I$, suppose $g : I \rightarrow J$ is n -times differentiable on a neighbourhood of x , and suppose $f : J \rightarrow \mathbb{R}$ is n -times differentiable on a neighbourhood of $g(x)$. Then $f \circ g$ is n -times differentiable at x , and

$$\frac{d^n}{dx^n} f(g(x)) = \sum_{\sum_{m=1}^n m k_m = n} \frac{n!}{\prod_{m=1}^n k_m! (m!)^{k_m}} f^{(k)}(g(x)) \prod_{m=1}^n (g^{(m)}(x))^{k_m},$$

where $k_m \geq 0$ and $k = \sum_{m=1}^n k_m$. Go to proof.

0.17 The Shape Questions

Definition 0.17.1 (Convex Function). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is *convex* on I if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Definition 0.17.2 (Concave Function). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is *concave* on I if $-f$ is convex on I .

Definition 0.17.3 (Inflection Point). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$ be an interior point. The point c is an *inflection point* of f if f changes convexity at c : there is a neighbourhood $(c - \delta, c + \delta) \subseteq I$ such that f is convex on one side of c and concave on the other side.

0.17.1 Interior Extrema

Definition 0.17.4 (Relative Minimum). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f has a *relative minimum* at c if there exists $\delta > 0$ such that

$$f(c) \leq f(x) \quad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(c) < f(x) \quad \text{for all } x \in (A \cap (c - \delta, c + \delta)) \setminus \{c\},$$

then f has a *strict relative minimum* at c .

Definition 0.17.5 (Relative Maximum). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f has a *relative maximum* at c if there exists $\delta > 0$ such that

$$f(x) \leq f(c) \quad \text{for all } x \in A \cap (c - \delta, c + \delta).$$

If, in addition,

$$f(x) < f(c) \quad \text{for all } x \in (A \cap (c - \delta, c + \delta)) \setminus \{c\},$$

then f has a *strict relative maximum* at c .

Definition 0.17.6 (Relative Extremum). Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f has a *relative extremum* at c if f has either a relative minimum at c or a relative maximum at c .

Theorem 0.17.7 (Interior Extremum Theorem). Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c and f is differentiable at c , then

$$f'(c) = 0.$$

Go to proof.

Theorem 0.17.8 (Necessary Condition for an Interior Extremum). Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c , then either $f'(c)$ does not exist, or

$$f'(c) = 0.$$

Go to proof.

Corollary 0.17.9 (Necessary Condition for an Extremum). Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c , then

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

Go to proof.

0.18 The Mean Value Theorems

Theorem 0.18.1 (Rolle's Theorem). Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$

Go to proof.

Theorem 0.18.2 (Mean Value Theorem). Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Go to proof.

Theorem 0.18.3 (Cauchy Mean Value Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f, g : [a, b] \rightarrow \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that*

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

If, in addition, $g'(x) \neq 0$ for every $x \in (a, b)$, then $g(b) \neq g(a)$ and

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Go to proof.

Corollary 0.18.4 (Derivative Bound Implies Lipschitz). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be differentiable on I . If there exists $M \geq 0$ such that*

$$|f'(x)| \leq M \quad \text{for every } x \in I,$$

then f is Lipschitz on I with constant M :

$$|f(x) - f(y)| \leq M|x - y| \quad \text{for every } x, y \in I.$$

Go to proof.

0.19 Reading the Graph from f' and f''

0.19.1 First-Order Shape

Definition 0.19.1 (Increasing at a Point). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f is increasing at c if there exists $\delta > 0$ such that for all*

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \leq f(c) \leq f(y)$.

Definition 0.19.2 (Decreasing at a Point). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, and let $c \in A$. We say that f is decreasing at c if there exists $\delta > 0$ such that for all*

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \geq f(c) \geq f(y)$.

Theorem 0.19.3 (Zero Derivative Implies Constant). *Let $I := [a, b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \rightarrow \mathbb{R}$ be continuous on I and differentiable on (a, b) . If*

$$f'(x) = 0 \quad \text{for all } x \in (a, b),$$

then f is constant on I . Go to proof.

Corollary 0.19.4 (Equal Derivatives Imply Difference Is Constant). *Let $I := [a, b] \subseteq \mathbb{R}$, and let $f, g : I \rightarrow \mathbb{R}$ be continuous on I and differentiable on (a, b) . If*

$$f'(x) = g'(x) \quad \text{for all } x \in (a, b),$$

then there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in I$. Go to proof.

Theorem 0.19.5 (Nondecreasing Iff Nonnegative Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ be differentiable on I . Then f is nondecreasing on I if and only if*

$$f'(x) \geq 0 \quad \text{for all } x \in I.$$

Go to proof.

Theorem 0.19.6 (Nonincreasing Iff Nonpositive Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ be differentiable on I . Then f is nonincreasing on I if and only if*

$$f'(x) \leq 0 \quad \text{for all } x \in I.$$

Go to proof.

Lemma 0.19.7 (Positive Derivative Implies Locally Increasing). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c . If $f'(c) > 0$, then there exists $\delta > 0$ such that*

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Lemma 0.19.8 (Negative Derivative Implies Locally Decreasing). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c . If $f'(c) < 0$, then there exists $\delta > 0$ such that*

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Theorem 0.19.9 (First Derivative Test: Relative Maximum). *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \rightarrow \mathbb{R}$ be continuous on I , and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b) . If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \geq 0 \quad \text{for all } x \in (c - \delta, c), \quad f'(x) \leq 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative maximum at c . Go to proof.

Theorem 0.19.10 (First Derivative Test: Relative Minimum). *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \rightarrow \mathbb{R}$ be continuous on I , and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b) . If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \leq 0 \quad \text{for all } x \in (c - \delta, c), \quad f'(x) \geq 0 \quad \text{for all } x \in (c, c + \delta),$$

then f has a relative minimum at c . Go to proof.

0.19.2 Second-Order Shape

0.19.3 Higher-Order Derivatives

Definition 0.19.11 (Second Derivative). *Let f be differentiable on a neighbourhood of c . If f' is differentiable at c , the *second derivative* of f at c is*

$$f''(c) := (f')'(c).$$

Definition 0.19.12 (Higher Derivatives). *Define $f^{(0)} := f$ and $f^{(1)} := f'$. For $n \geq 2$, if $f^{(n-1)}$ is differentiable at c , define*

$$f^{(n)}(c) := (f^{(n-1)})'(c).$$

0.19.4 Second-Order Shape

Theorem 0.19.13 (Second-Derivative Convexity Test). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be twice differentiable on I . If*

$$f''(x) \geq 0 \quad \text{for all } x \in I,$$

then f is convex on I . Go to proof.

Theorem 0.19.14 (Second-Derivative Concavity Test). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be twice differentiable on I . If*

$$f''(x) \leq 0 \quad \text{for all } x \in I,$$

then f is concave on I . Go to proof.

Theorem 0.19.15 (Second Derivative Test). *Let f be twice differentiable in a neighbourhood of c , and suppose $f'(c) = 0$.*

- *If $f''(c) > 0$, then f has a strict relative minimum at c .*
- *If $f''(c) < 0$, then f has a strict relative maximum at c .*

Go to proof.

Proposition 0.19.16 (Inflection Point Necessary Condition). *Let f have an inflection point at c . If f'' exists on a neighbourhood of c and is continuous at c , then*

$$f''(c) = 0.$$

Go to proof.

0.19.5 Darboux's Theorem

Theorem 0.19.17 (Darboux's Theorem). *Let $I := [a, b] \subseteq \mathbb{R}$ and let $f : I \rightarrow \mathbb{R}$ be differentiable on I . If k is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a, b)$ such that*

$$f'(c) = k.$$

Go to proof.

0.19.6 Smoothness Classes

Definition 0.19.18 (Continuously Differentiable; Class C^1). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is continuously differentiable on I , written $f \in C^1(I)$, if f is differentiable at every point of I and the derivative $f' : I \rightarrow \mathbb{R}$ is continuous on I .*

Definition 0.19.19 (Class C^k). *Let $I \subseteq \mathbb{R}$ be an interval, let $k \in \mathbb{N}$, and let $f : I \rightarrow \mathbb{R}$. The function f is of class C^k on I , written $f \in C^k(I)$, if the derivatives $f^{(0)}, f^{(1)}, \dots, f^{(k)}$ exist on I and $f^{(k)}$ is continuous on I . By convention, $C^0(I)$ is the class of continuous functions on I .*

Definition 0.19.20 (Smooth Function; Class C^∞). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is smooth, or of class C^∞ , written $f \in C^\infty(I)$, if $f \in C^k(I)$ for every $k \in \mathbb{N}$. Equivalently,*

$$C^\infty(I) = \bigcap_{k=0}^{\infty} C^k(I).$$

Definition 0.19.21 (Real-Analytic Function; Class C^ω). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is real-analytic, or of class C^ω , written $f \in C^\omega(I)$, if for every $a \in I$ there exists $r > 0$ such that*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad \text{for every } x \in (a-r, a+r) \cap I.$$

Theorem 0.19.22 (The Smoothness Tower). *For every nondegenerate interval I , the inclusions*

$$C^0(I) \supseteq C^1(I) \supseteq C^2(I) \supseteq \cdots \supseteq C^\infty(I) \supseteq C^\omega(I)$$

hold, and each displayed inclusion is strict. Go to proof.

Definition 0.19.23 (Lipschitz-Derivative Class $C^{1,1}$). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$. The function f is of class $C^{1,1}$ on I , written $f \in C^{1,1}(I)$, if $f \in C^1(I)$ and f' is Lipschitz on I : there exists $L \geq 0$ such that

$$|f'(x) - f'(y)| \leq L|x - y| \quad \text{for all } x, y \in I.$$

Theorem 0.19.24 (Placement of $C^{1,1}$). *For every nondegenerate interval I ,*

$$C^{1,1}(I) \subsetneq C^1(I),$$

and $C^{1,1}(I)$ is not contained in $C^2(I)$. If K is a compact interval, then

$$C^2(K) \subseteq C^{1,1}(K) \subsetneq C^1(K),$$

and the first inclusion is strict. Go to proof.

Corollary 0.19.25 (Bounded Second Derivative Implies $C^{1,1}$). *Let $I \subseteq \mathbb{R}$ be an interval and let $f \in C^2(I)$. If $|f''(x)| \leq M$ for all $x \in I$, then $f \in C^{1,1}(I)$, and f' is Lipschitz with constant M . Go to proof.*

0.20 The Algebra of Derivatives

0.20.1 From Characterization to Computation

Theorem 0.20.1 (Constant Multiple Rule). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, let $c \in A$ be a limit point of A , and let $\alpha \in \mathbb{R}$. If f is differentiable at c , then αf is differentiable at c , and*

$$(\alpha f)'(c) = \alpha f'(c).$$

Go to proof.

Theorem 0.20.2 (Sum Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . If f and g are differentiable at c , then $f + g$ is differentiable at c , and*

$$(f + g)'(c) = f'(c) + g'(c).$$

Go to proof.

Theorem 0.20.3 (Product Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . If f and g are differentiable at c , then fg is differentiable at c , and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Go to proof.

Theorem 0.20.4 (Quotient Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . Suppose that f and g are differentiable at c and that $g(c) \neq 0$. Then f/g is differentiable at c , and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Go to proof.

0.20.2 Closure

Corollary 0.20.5 (Finite Sum Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \rightarrow \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A . If each f_i is differentiable at c , then $\sum_{i=1}^n \alpha_i f_i$ is differentiable at c , and*

$$\left(\sum_{i=1}^n \alpha_i f_i \right)'(c) = \sum_{i=1}^n \alpha_i f'_i(c).$$

Go to proof.

Corollary 0.20.6 (Extended Product Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \rightarrow \mathbb{R}$, and let $c \in A$ be a limit point of A . If each f_i is differentiable at c , then $\prod_{i=1}^n f_i$ is differentiable at c , and*

$$\left(\prod_{i=1}^n f_i \right)'(c) = \sum_{k=1}^n f'_k(c) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(c).$$

Go to proof.

Corollary 0.20.7 (Power Rule for Integer Powers of a Function). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$, let $c \in A$ be a limit point of A , and let $n \in \mathbb{N}$. If f is differentiable at c , then f^n is differentiable at c , and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

Go to proof.

Theorem 0.20.8 (Finite Linear Combination Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \rightarrow \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A . If each f_i is differentiable at c , then*

$$\left(\sum_{i=1}^n \alpha_i f_i \right)'(c) = \sum_{i=1}^n \alpha_i f'_i(c).$$

Go to proof.

0.20.3 Global Forms on an Interval

Theorem 0.20.9 (Constant Multiple Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If f is differentiable on I , then αf is differentiable on I , and*

$$(\alpha f)'(x) = \alpha f'(x) \quad \text{for all } x \in I.$$

Go to proof.

Theorem 0.20.10 (Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \rightarrow \mathbb{R}$. If f and g are differentiable on I , then $f + g$ is differentiable on I , and*

$$(f + g)'(x) = f'(x) + g'(x) \quad \text{for all } x \in I.$$

Go to proof.

Theorem 0.20.11 (Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \rightarrow \mathbb{R}$. If f and g are differentiable on I , then fg is differentiable on I , and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \quad \text{for all } x \in I.$$

Go to proof.

Theorem 0.20.12 (Quotient Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \rightarrow \mathbb{R}$. Suppose that f and g are differentiable on I and that $g(x) \neq 0$ for all $x \in I$. Then f/g is differentiable on I , and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \quad \text{for all } x \in I.$$

Go to proof.

Corollary 0.20.13 (Finite Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \rightarrow \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I , then $\sum_{i=1}^n \alpha_i f_i$ is differentiable on I , and*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(x) = \sum_{i=1}^n \alpha_i f'_i(x) \quad \text{for all } x \in I.$$

Go to proof.

Corollary 0.20.14 (Extended Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \dots, f_n : I \rightarrow \mathbb{R}$. If each f_i is differentiable on I , then $\prod_{i=1}^n f_i$ is differentiable on I , and*

$$\left(\prod_{i=1}^n f_i\right)'(x) = \sum_{k=1}^n f'_k(x) \prod_{\substack{i=1 \\ i \neq k}}^n f_i(x) \quad \text{for all } x \in I.$$

Go to proof.

Corollary 0.20.15 (Power Rule for Integer Powers of a Function on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$, and let $n \in \mathbb{N}$. If f is differentiable on I , then f^n is differentiable on I , and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \quad \text{for all } x \in I.$$

Go to proof.

Theorem 0.20.16 (Finite Linear Combination Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \rightarrow \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I , then*

$$\left(\sum_{i=1}^n \alpha_i f_i\right)'(x) = \sum_{i=1}^n \alpha_i f'_i(x) \quad \text{for all } x \in I.$$

Go to proof.

0.20.4 Inverting the Operation

Theorem 0.20.17 (Inverse Function Theorem, One Variable). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \rightarrow \mathbb{R}$ be of class C^1 , and let $c \in I$ satisfy $f'(c) \neq 0$. Then there exist open intervals $U \subseteq I$ and $V \subseteq \mathbb{R}$, with $c \in U$ and $f(c) \in V$, such that:*

1. $f|_U : U \rightarrow V$ is a bijection;
2. the inverse $g = (f|_U)^{-1} : V \rightarrow U$ is of class C^1 ;
3. for every $y \in V$,

$$g'(y) = \frac{1}{f'(g(y))}.$$

Go to proof.

Corollary 0.20.18 (Derivative of the Inverse Function). *Under the hypotheses of the one-variable Inverse Function Theorem, the local inverse g satisfies*

$$g' = \frac{1}{f' \circ g}$$

on V . More globally, if $f : I \rightarrow J$ is a C^1 bijection with $f'(x) \neq 0$ for every $x \in I$, then $f^{-1} : J \rightarrow I$ is C^1 and

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))} \quad \text{for every } y \in J.$$

Go to proof.

0.20.5 Indeterminate Forms: L'Hôpital's Rule

Theorem 0.20.19 (L'Hôpital's Rule, 0/0 Form). *Let $a < b$, and let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

The analogous statements at left-hand endpoints, at two-sided interior points, and at infinity follow by the same reductions. Go to proof.

Theorem 0.20.20 (L'Hôpital's Rule, ∞/∞ Form). *Let $a < b$, and let $f, g : (a, b) \rightarrow \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose*

$$\lim_{x \rightarrow a^+} |g(x)| = +\infty \quad \text{and} \quad \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)} = L$$

for $L \in \mathbb{R} \cup \{-\infty, +\infty\}$. Then

$$\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)} = L.$$

No separate hypothesis on $\lim_{x \rightarrow a^+} f(x)$ is required. Go to proof.

0.21 Taylor: Construction and Its Error

Definition 0.21.1 (Taylor Polynomial at a Point). Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \rightarrow \mathbb{R}$ have derivatives $f^{(k)}(a)$ for $0 \leq k \leq n$. The **Taylor polynomial of order n** for f at a is

$$T_{n,a}f(x) := \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

Definition 0.21.2 (Taylor Remainder). Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \rightarrow \mathbb{R}$ have a Taylor polynomial $T_{n,a}f$. The **Taylor remainder of order n** at a is the function $R_{n,a}f : I \rightarrow \mathbb{R}$ defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

Definition 0.21.3 (Maclaurin Polynomial). Let $n \in \mathbb{N}$, and let f be defined on an interval containing 0 with derivatives $f^{(k)}(0)$ for $0 \leq k \leq n$. The **Maclaurin polynomial of order n** for f is

$$T_{n,0}f(x) := \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k.$$

Theorem 0.21.4 (Taylor's Theorem with Lagrange Remainder). *Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a, b] \rightarrow \mathbb{R}$. Suppose that $f, f', \dots, f^{(n)}$ are continuous on $[a, b]$ and that $f^{(n+1)}$ exists on (a, b) . Then, for every $x \in (a, b)$, there exists c strictly between a and x such that*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}.$$

Go to proof.

Corollary 0.21.5 (Taylor Expansion with Peano Remainder). *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \rightarrow \mathbb{R}$ have derivatives up to order n in a neighbourhood of a . If $f^{(n)}$ is continuous at a , then*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + o((x-a)^n) \quad \text{as } x \rightarrow a.$$

Go to proof.

Corollary 0.21.6 (First-Order Peano Remainder). *Let f be differentiable at c . Then, as $h \rightarrow 0$,*

$$f(c+h) = f(c) + f'(c)h + o(h).$$

Go to proof.

Proposition 0.21.7 (The Flat Function: Smooth but Not Analytic). *Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by*

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then:

1. $f \in C^\infty(\mathbb{R})$;
2. $f^{(n)}(0) = 0$ for every $n \geq 0$;
3. the Maclaurin series of f is identically 0, so $f \notin C^\omega$ on any neighbourhood of 0.

In particular, $C^\infty(\mathbb{R}) \setminus C^\omega(\mathbb{R}) \neq \emptyset$. Go to proof.

0.21.1 The Differential

Definition 0.21.8 (Differentiability by a Differential). *Let f be defined on a neighbourhood of $c \in \mathbb{R}$. The function f is differentiable at c in the differential sense if there exists a linear map $L : \mathbb{R} \rightarrow \mathbb{R}$ such that*

$$f(c+h) - f(c) = L(h) + o(h) \quad \text{as } h \rightarrow 0.$$

In that case the differential of f at c is $df_c := L$.

Theorem 0.21.9 (Differential and Derivative Agree). *Let f be defined on a neighbourhood of $c \in \mathbb{R}$. Then f is differentiable at c in the ordinary derivative sense if and only if it is differentiable at c in the differential sense. When this holds,*

$$df_c(h) = f'(c)h.$$

Go to proof.

Theorem 0.21.10 (Uniqueness of the Differential). *If $L_1, L_2 : \mathbb{R} \rightarrow \mathbb{R}$ are linear maps and*

$$f(c+h) - f(c) = L_1(h) + o(h) = L_2(h) + o(h) \quad \text{as } h \rightarrow 0,$$

then $L_1 = L_2$. Go to proof.

Theorem 0.21.11 (Differential Continuity Criterion). *If f is differentiable at c in the differential sense, then f is continuous at c . Go to proof.*

Theorem 0.21.12 (Chain Rule for Differentials). *If f is differentiable at c and g is differentiable at $f(c)$, then*

$$d(g \circ f)_c = dg_{f(c)} \circ df_c.$$

Go to proof.

Theorem 0.21.13 (Linearity of the Differential). *If f and g are differentiable at c and $\alpha, \beta \in \mathbb{R}$, then*

$$d(\alpha f + \beta g)_c = \alpha df_c + \beta dg_c.$$

Go to proof.

Proofs